Energy-Efficient and Near Real-Time Routing for Disaster Monitoring in Wireless Sensor Networks

Graph-Attentive Policy Fusion (GAPF)

Georges Parfait Djimefo Kapen

Director: **Pr. Ranwa Al Mallah**

Co-director: **Pr. Samuel Pierre**, Senior Member, IEEE

Mobile Computing and Networking Research Laboratory (LARIM)
Dept. of Computer and Software Engineering
Polytechnique Montréal

PolyCongRÈs 2026

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Outline

Why Disaster Monitoring Matters

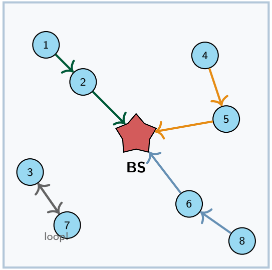
A Real-World Crisis

In **2023**: **398** natural disasters, **95 000** deaths, **$380 B** in losses

Source: EEA, Statista 2023

Wireless Sensor Networks (WSNs) are essential tools:

- Self-configuring, infrastructure-independent
- Accurate data with minimal human intervention
- Resilient when infrastructure is destroyed



Multi-hop routing to Base Station

Core tension: Deliver all data reliably *vs.* Preserve limited battery energy

Limitations of Current Approaches

Metaheuristic Methods (PSO, Firefly, GWO...)

- Assume **static** sensor positions
- No adaptation to topology changes
- PDR **collapses** $<$**10%** at scale

Single-Agent RL (Q-learning)

- Reward $=$ simple **weighted sum**
- Cannot capture multi-objective complexity

Multi-Agent RL (QMIX, QTRAN)

- QMIX: limited by **monotonicity** constraint
- QTRAN: better theory, **harder to train**
- Both **ignore network topology**
- **Memory explosion**: up to **13.8 GB**

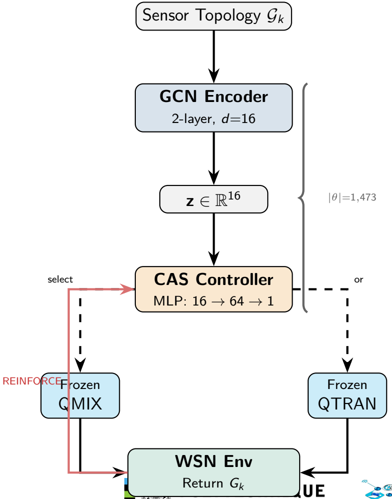
Our Goal

Design a **lightweight, topology-aware** meta-controller that selects the best routing expert while satisfying **energy, latency, and memory** constraints.

GAPF: Graph-Attentive Policy Fusion — Overview

Three key contributions:

1. **GAPF Meta-Controller**
   GCN encodes WSN topology → CAS selects best expert per episode. Only **1 473 params**.
2. **Monotonic Attention Reward**
   Learned attention scalarizes 4 objectives with **monotonicity guarantee**. Tuned by **Evolution Strategies**.
3. **Realistic Gym Environment**
   Dynamic sensor mobility, energy-aware multi-hop, custom OpenAI Gym.



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Bi-Level Reward Architecture

Inner level: 4-dim reward vector per hop

$$\mathbf{r}_i^{(t)} = [\underbrace{r_1}_{\text{angle}}, \underbrace{r_2}_{\text{energy}}, \underbrace{r_3}_{\text{balance}}, \underbrace{r_4}_{\text{load}}] \in [0,1]^4$$

Attention weighting (learned, input-independent):

$$\alpha_j = \frac{\exp(\mathbf{k}_j^\top \mathbf{q}/\sqrt{d})}{\sum_{l=1}^{4} \exp(\mathbf{k}_l^\top \mathbf{q}/\sqrt{d})}$$

Monotonic network ($4 \to 16 \to 16 \to 1$): $\mathbf{W}_\ell \geq 0$ via softplus

$\Rightarrow$ **Pareto-dominating action always gets higher reward**

Outer level: Evolution Strategies

Meta-objective after each episode:

$$J = \underbrace{1 - \frac{\sum (E_0 - e_i^{\text{rem}})}{N \cdot E_0}}_{\text{energy eff.}} + \underbrace{1 - \frac{\sigma(\mathbf{e}_{\text{rem}})}{E_0/2}}_{\text{energy bal.}}$$

$\rightarrow$ Decouples reward shaping from RL gradient

$\rightarrow$ Adapts weights to each topology

Why is this novel?

- No manual weight tuning
- Mathematically guaranteed monotonicity
- Generalizes across topologies

Results: Packet Delivery & Energy Efficiency

7 scenarios × 3 seeds × 1 000 episodes = **21 000 test episodes**

Packet Delivery Ratio (PDR)

n	R	QMIX	QTRAN	**GAPF**
20	25	.559	.558	**.753**
30	25	.568	.567	**.877**
50	35	.553	.552	**.990**
70	35	.545	.545	**.980**
100	35	.464	.462	**.980**
100	50	.669	.684	**.980**

+43% to +111% relative improvement

Energy Balance (σ_E, ↓ better)

n	R	QMIX	QTRAN	**GAPF**
20	25	.0075	.0075	**.0043**
30	25	.0111	.0112	**.0035**
50	35	.0264	.0265	**.0039**
70	35	.0318	.0319	**.0049**
100	35	.0455	.0457	**.0061**
100	50	.0559	.0559	**.0061**

3–9× better energy balance

QMIX ≈ QTRAN (<2% gap): gain comes from **topology-aware fusion**, not from expert choice alone

Deployment Feasibility: From Server to Sensor

Feasibility Compliance

Metric	QMIX	QTRAN	GAPF
Peak RAM	3.2–13.8 GB		**684–951 MB**
Mem <1 GB?	FAIL	FAIL	PASS
Latency p_{95}	0.27 ms	0.17 ms	**0.03 ms**

CTDE Paradigm

- **Training:** heavy components on server
- **Deployment:** only lightweight GRU on each sensor
- Mixers, reward network, ES loop $\rightarrow$ **discarded**

What Runs on the Sensor?

Resource	Requirement
Flash (weights)	$\sim$100 KB
SRAM (runtime)	$\sim$2 KB
Compute/step	$\sim$29K MACs
Inference time	<1 ms

Compatible with:

STM32L4 (1 MB flash, 256 KB SRAM)

nRF52840 (1 MB flash, 256 KB RAM)

GAPF adds **zero overhead** on sensor nodes

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Real-World Impact

♡ Life-Saving Responsiveness

- $\geq$**98% packet delivery** at all scales
- Every undelivered reading = **blind spot**
- Near real-time: $<$1 ms per routing decision

★ Green AI

- **2–3.5**$\times$ less energy $\Rightarrow$ longer lifetime
- Fewer batteries in disaster zones; less e-waste
- Only **1 473 parameters** to train

▶ Broader Applicability

"Lightweight policy fusion > monolithic models"

- Vehicular ad-hoc networks (VANETs)
- Drone swarm coordination
- Smart grid load balancing

Responsible AI

- **Interpretable**: binary selection is auditable
- **Efficient**: minimal training compute
- **Deployable**: no GPU at inference

Conclusion & Future Work

Summary of Key Results

Achievement	**Result**
Packet Delivery Ratio	$\geq$**98%** (20–100 sensors)
PDR improvement vs. baselines	**+43% to +111%**
Energy reduction	**2–3.5**$\times$ less than QMIX/QTRAN
Energy balance	**3–9**$\times$ better
Memory reduction	**13**$\times$ less (951 MB vs. 13.2 GB)
Trainable parameters	Only **1 473**
Feasibility constraints	**ALL PASSED**

Future directions:

1. Hardware-in-the-loop on nRF52840 motes
2. Cross-domain: flood, wildfire, earthquake monitoring
3. Dynamic step-level expert fusion
4. Multi-expert generalization ($>$2 experts)

Thank You!

Questions?

Georges Parfait Djimefo Kapen

Director: Pr. Ranwa Al Mallah | Co-director: Pr. Samuel Pierre

georges-parfait.djimefo-kapen@polymtl.ca

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